Meta-Analysis_TMS_Physical_Activity

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R Markdown file set-up and packages

Packages required: 1) dmetar 2) tidyverse 3) meta 4) metafor 5) esc 6) TOSTER

Meta-analysis

All analyses were conducted in the R statistical environment (R Core Team, 2023) using the meta (Schwarzer, 2023) and metafor (Viechtbauer, 2010, 2023) packages.

Pearson product–moment correlations from eligible studies were pooled to examine the relationship between physical activity and transcranial magnetic stimulation (TMS) measures of cortico-spinal excitability, cortical inhibition, and cortical facilitation. Prior to analysis, correlations were Fisher's z-transformed to stabilize variance, and the corresponding sampling variances were calculated. Pooled estimates were then converted back to Pearson correlations for ease of interpretation.

Meta-analyses were conducted using multilevel random-effects models implemented with the metafor R package (Viechtbauer, 2010, 2023). This approach accounts for statistical dependence among multiple correlation estimates reported within the same study. A multilevel approach was used because several studies contributed more than one correlation estimate (e.g., different TMS outcomes or measurement conditions), which introduces statistical dependence among effect sizes. Multilevel meta-analysis appropriately accounts for this dependency without requiring aggregation of effect sizes within studies. Specifically, correlation estimates were nested within studies, resulting in a three-level model (level 1: sampling variance, level 2: correlation estimates within studies, level 3: studies). Between-study heterogeneity was estimated using the restricted maximum likelihood (REML) estimator (Viechtbauer, 2005). Heterogeneity was quantified using the I^2 statistic, which represents the proportion of total variability in observed effect sizes that is not attributable to sampling error (Higgins & Thompson, 2002). Because a multilevel meta-analytic model was used, I^2 values were decomposed across variance components to estimate the proportion of variability attributable to differences between studies and between effect sizes within studies.

To reduce the risk of inflated type I error rates, confidence intervals for pooled estimates were calculated using a Knapp–Hartung adjustment (Knapp & Hartung, 2003). In addition to pooled correlations and their confidence intervals, prediction intervals were calculated to estimate the range within which the true effect size of a future study is expected to fall based on the current evidence. Unlike confidence intervals, which describe the precision of the pooled estimate, prediction intervals incorporate between-study heterogeneity and therefore reflect the expected variability of effects across future studies.

For visualization purposes, forest plots were produced using the meta R package (Schwarzer, 2023) with a RevMan-style layout. These plots were based on random-effects models equivalent to the multilevel analyses and were used solely for graphical presentation. Sensitivity analyses using conventional random-effects models were also conducted using the metacor function in the meta package.

Publication bias was visually assessed using contour-enhanced funnel plots, which display the Fisher's z-transformed correlations against their standard errors. In the absence of publication bias, studies are expected to form a symmetrical inverted funnel around the pooled effect size.

Subgroup analyses were conducted to examine whether pooled correlations differed across types of inhibitory modulation (SICI, LICI, cSP, and iSP) and facilitatory modultation (ICF and SICF)

Import Pearson r data

Data file glimpse

Calculating z and var.z

Excitability (CSE): Primary analysis of using metafor::rma.mv()

```
m.cor.excitability <- rma.mv(yi = z, # pool correlation coefficients Fishers Z transformation https://b
                 V = var.z, # sampling variance of Fisher's Z values: 1/(N-3), where N = the sample size
                 slab = author,
                 data = excitability_r,
                 random = ~ 1 | author/cor_id,
                 test = "t",
                 method = "REML")
summary(m.cor.excitability)
```

```
##
```

```
## Multivariate Meta-Analysis Model (k = 15; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   0.2075   -0.4149    5.5851    7.5022    7.9851
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000     14     no         author
## sigma^2.2  0.0060  0.0772     15     no  author/cor_id
##
## Test for Heterogeneity:
## Q(df = 14) = 18.5446, p-val = 0.1831
##
## Model Results:
##
## estimate      se     tval  df    pval    ci.lb   ci.ub
##  -0.0063  0.0548  -0.1147  14  0.9103  -0.1239  0.1113
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Excitability (CSE): Converting Fisher's z back to Pearson's r

```r
round(convert_z2r(-0.0063), 2) # point estimate
```

```
## [1] -0.01
```

```r
round(convert_z2r(-0.1239), 2) # lower CI
```

```
## [1] -0.12
```

```r
round(convert_z2r(0.1113), 2) # Upper CI
```

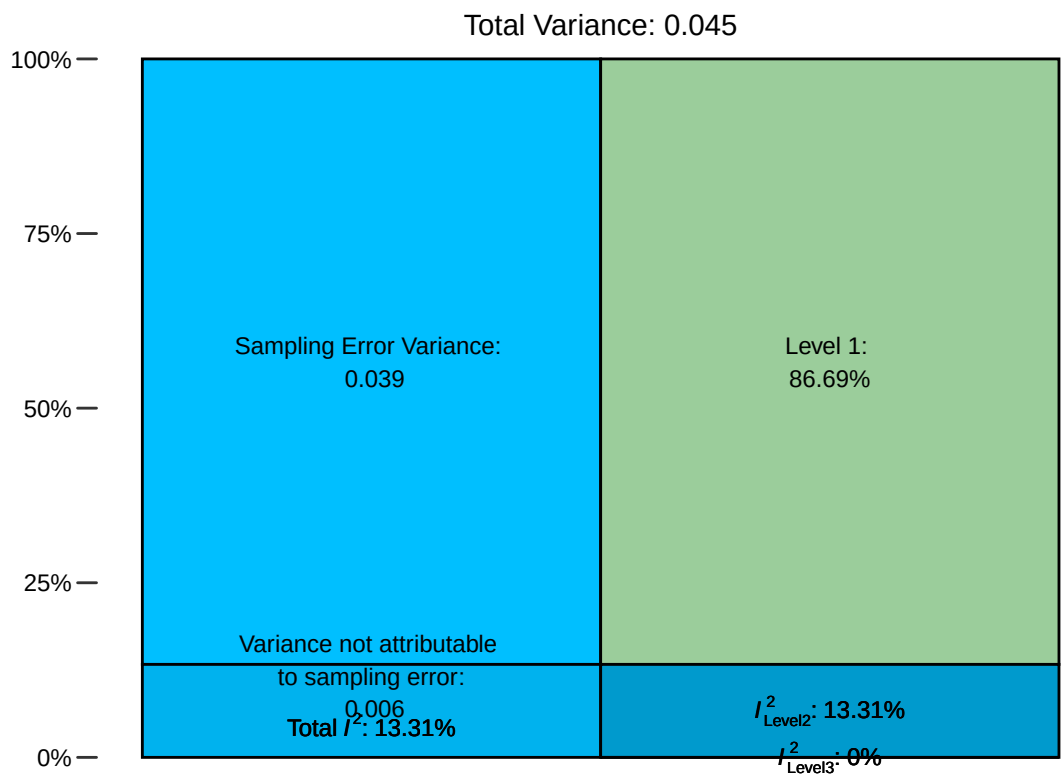
```
## [1] 0.11
```

Excitability (CSE): Explore heterogeneity

```r
i2 <- var.comp(m.cor.excitability)
summary(i2)
```

```
##         % of total variance    I2
## Level 1        8.668799e+01   ---
## Level 2        1.331201e+01 13.31
## Level 3        1.616037e-09     0
## Total I2: 13.31%
```

```r
plot(i2)
```



ADD TEXT

Excitability (CSE): Sensitivity analysis using meta::metacor()

```r
m.excitability <- metacor(cor = cor,
              n = n,
              studlab = author,
              data = excitability_r,
              cluster = cluster,
              common = FALSE,
              random = TRUE,
              method.tau = "REML",
              method.random.ci = "HK",
              title = "Sensitivity meta-analysis using metacor")

m.excitability
```

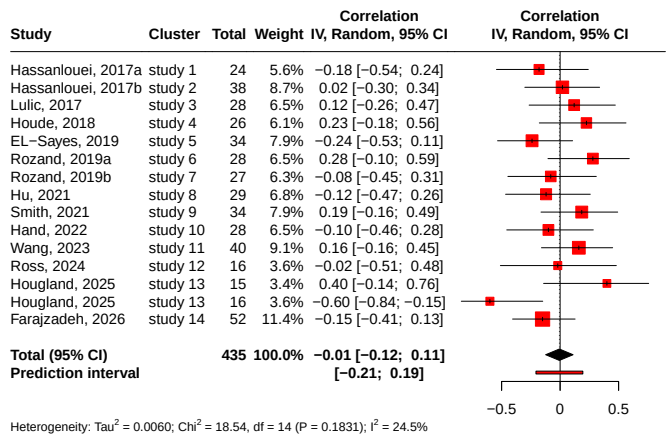
```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 14
## Number of estimates: k = 15
```

```
## Number of observations: o = 435
##
##                           COR            95%-CI     t p-value
## Random effects model -0.0063 [-0.1233; 0.1109] -0.11  0.9103
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 < 0.0001 [0.0000; 0.0391]; tau.1 < 0.0001 [0.0000; 0.1978] (between cluster)
##  tau^2.2 = 0.0060 [0.0000; 0.0761]; tau.2 = 0.0772 [0.0000; 0.2759] (within cluster)
##  I^2 = 24.5% [0.0%; 59.2%]; H = 1.15 [1.00; 1.57]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  18.54   14  0.1831
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 14)
## - Fisher's z transformation of correlations
```

#Corticospinal excitability A total of 15 correlation estimates from 14 studies (435 participants) were included in the meta-analysis examining the association between habitual physical activity and corticospinal excitability. The multilevel random-effects meta-analysis indicated no significant association between physical activity and corticospinal excitability (r = −0.01, 95% CI [−0.12, 0.11], p = 0.91). The prediction interval ranged from −0.21 to 0.19, suggesting that the true effect in future studies is likely to fall within this range. Between-study heterogeneity was low, with a total I^2 of 13.3%, indicating that most of the observed variability was attributable to sampling error rather than true differences between studies. Decomposition of variance components indicated that 13.31% of the total variability occurred between effect sizes within studies, whereas between-study heterogeneity was negligible. To further evaluate whether the association could be considered practically negligible, a two one-sided tests (TOST) equivalence test was conducted using equivalence bounds corresponding to a small effect size (−0.10 to 0.10). The equivalence test was significant (Z = 1.72, p = 0.043), and the 90% confidence interval (−0.096 to 0.084) fell entirely within the predefined equivalence bounds, supporting the conclusion that the association between physical activity and corticospinal excitability is negligible. Sensitivity analyses using a conventional random-effects model implemented with the metacor function produced virtually identical results (r = −0.01, 95% CI [−0.12, 0.11], p = 0.91), supporting the robustness of the findings.

Excitability (CSE): Forest plot

```r
forest(m.excitability,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Excitability (CSE): Save plot

```
library(svglite)

svglite("forest_plot_excitability.svg", height = 7, width = 7)

forest(m.excitability,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Inhibition (SICI, LICI, cSP, iSP): Primary analysis of using metafor::rma.mv()

```
m.cor.inhibition <- rma.mv(yi = z, # pool correlation coefficients Fishers Z transformation https://boo
                V = var.z, # sampling variance of Fisher's Z values: 1/(N-3), where N = the sample size
                slab = author,
                data = inhibition_r,
                random = ~ 1 | author/cor_id,
                test = "t",
                method = "REML")
summary(m.cor.inhibition)
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   2.6878   -5.3756    0.6244    2.3192    3.2910
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0066  0.0814     11     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Heterogeneity:
## Q(df = 13) = 14.5843, p-val = 0.3340
##
## Model Results:
##
## estimate      se    tval  df    pval   ci.lb   ci.ub
##   0.1642  0.0562  2.9241  13  0.0118  0.0429  0.2856  *
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Inhibition (SICI, LICI, cSP, iSP): Converting Fisher's z back to Pearson's r

```r
round(convert_z2r(0.1642), 2) # point estimate
```

```
## [1] 0.16
```

```r
round(convert_z2r(0.0429), 2) # lower CI
```

```
## [1] 0.04
```

```r
round(convert_z2r(0.2856), 2) # Upper CI
```

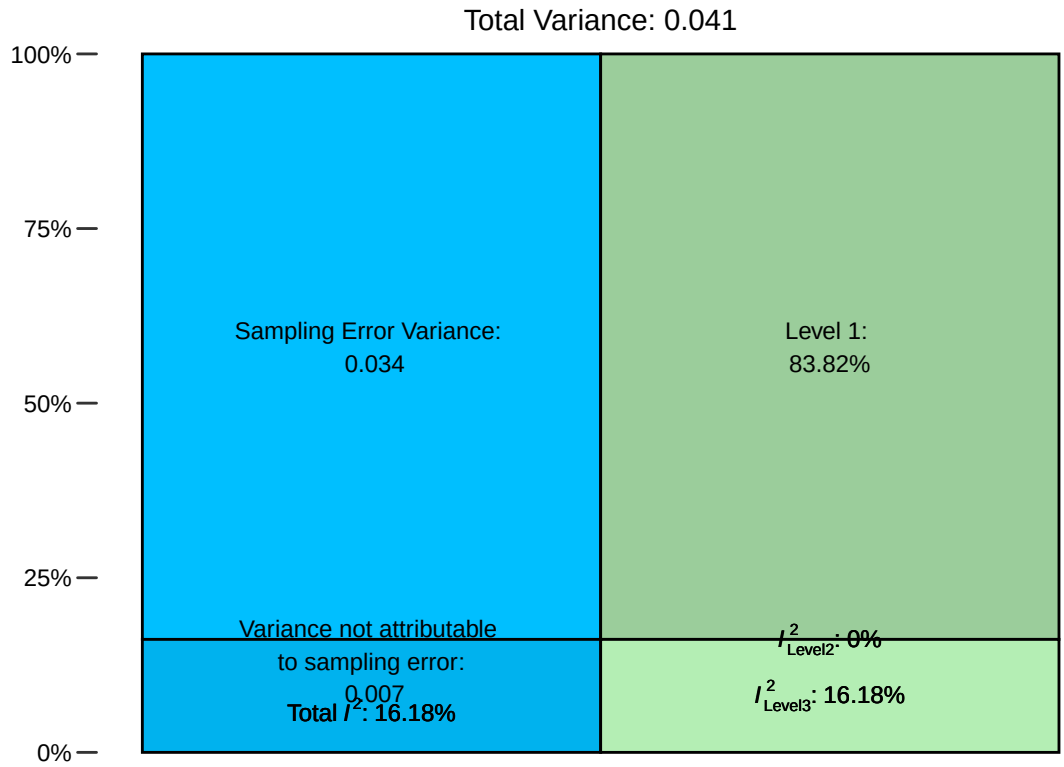
```
## [1] 0.28
```

Inhibition (SICI, LICI, cSP, iSP): Explore heterogeneity

```r
i2 <- var.comp(m.cor.inhibition)
summary(i2)
```

```
##         % of total variance    I2
## Level 1        8.382294e+01   ---
## Level 2        8.074525e-08     0
## Level 3        1.617706e+01 16.18
## Total I2: 16.18%
```

```r
plot(i2)
```



ADD TEXT

Inhibition (SICI, LICI, cSP, iSP): Sensitivity analysis using meta::metacor()

```r
m.inhibition <- metacor(cor = cor,
              n = n,
              studlab = author,
              data = inhibition_r,
              cluster = cluster,
              common = FALSE,
              random = TRUE,
              method.tau = "REML",
              method.random.ci = "HK",
              title = "Sensitivity meta-analysis using metacor")

m.inhibition
```

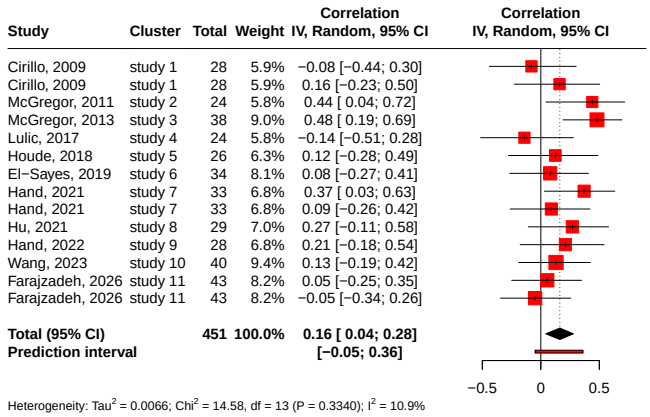
```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 11
## Number of estimates: k = 14
```

```
## Number of observations: o = 451
##
##                          COR           95%-CI    t p-value
## Random effects model 0.1628 [0.0429; 0.2780] 2.92  0.0118
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0066 [0.0000; 0.0694]; tau.1 = 0.0814 [0.0000; 0.2634] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0572]; tau.2 < 0.0001 [0.0000; 0.2392] (within cluster)
##  I^2 = 10.9% [0.0%; 49.2%]; H = 1.06 [1.00; 1.40]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  14.58   13  0.3340
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 13)
## - Fisher's z transformation of correlations
```

#Cortical inhibition A total of 14 correlation estimates from 11 studies (451 participants) were included in the meta-analysis examining the association between habitual physical activity and measures of cortical inhibition (SICI, LICI, cSP, and iSP). The multilevel random-effects meta-analysis indicated a significant positive association between physical activity and cortical inhibition (r = 0.16, 95% CI [0.04, 0.28], p = 0.012). This effect corresponds to a small positive correlation, suggesting that higher levels of habitual physical activity are associated with greater cortical inhibitory processes. Between-study heterogeneity was low, with a total I^2 of 16.18%, indicating that most of the variability in observed correlations was attributable to sampling error. Variance decomposition showed that 16.18% of the variability occurred between studies, while no meaningful heterogeneity was observed between effect sizes within studies. Sensitivity analyses using a conventional random-effects model implemented with the metacor function yielded virtually identical results (r = 0.16, 95% CI [0.04, 0.28], p = 0.012), supporting the robustness of the findings. The prediction interval ranged from −0.05 to 0.36, indicating that although the overall association was positive, future studies could observe effects ranging from negligible to moderately positive. Subgroup analyses examining differences across inhibitory mechanisms (SICI, LICI, cSP, and iSP) revealed no statistically significant differences between modulation types (Q = 2.95, df = 3, p = 0.085). Nevertheless, exploratory estimates suggested that the association was strongest for interhemispheric inhibition (iSP) (r 0.47) and smaller for SICI and cSP, while the estimate for LICI was close to zero.

Inhibition (SICI, LICI, cSP, iSP): Forest plot

```r
forest(m.inhibition,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Inhibition (SICI, LICI, cSP, iSP): Save plot

```
library(svglite)

svglite("forest_plot_inhibition.svg", height = 7, width = 7)

forest(m.inhibition,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Facilitation (ICF, SICF): Primary analysis of using metafor::rma.mv()

```
m.cor.facilitation <- rma.mv(yi = z, # pool correlation coefficients Fishers Z transformation https://b
                V = var.z, # sampling variance of Fisher's Z values: 1/(N-3), where N = the sample size
                slab = author,
                data = facilitation_r,
                random = ~ 1 | author/cor_id,
                test = "t",
                method = "REML")
summary(m.cor.facilitation)
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   2.6971   -5.3942    0.6058    2.3007    3.2725
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000      8     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Heterogeneity:
## Q(df = 13) = 12.8052, p-val = 0.4630
##
## Model Results:
##
## estimate      se    tval  df    pval    ci.lb   ci.ub
##   0.0674  0.0520  1.2971  13  0.2172  -0.0449  0.1797
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Facilitation (ICF, SICF): Converting Fisher's z back to Pearson's r

```r
round(convert_z2r(0.0674), 2) # point estimate
```

```
## [1] 0.07
```

```r
round(convert_z2r(-0.0449), 2) # lower CI
```

```
## [1] -0.04
```

```r
round(convert_z2r(0.1797), 2) # Upper CI
```

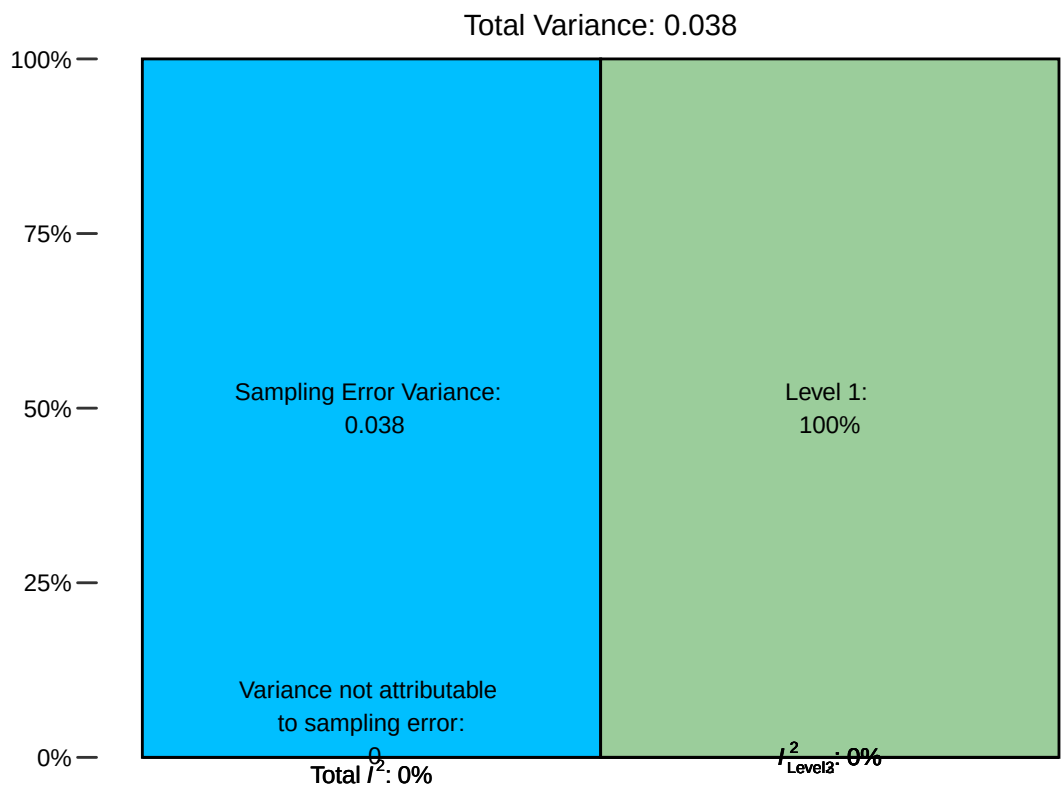
```
## [1] 0.18
```

Facilitation (ICF, SICF): Explore heterogeneity

```r
i2 <- var.comp(m.cor.facilitation)
summary(i2)
```

```
##         % of total variance  I2
## Level 1         1.00000e+02 ---
## Level 2         2.13181e-08   0
## Level 3         0.00000e+00   0
## Total I2: 0%
```

```r
plot(i2)
```



ADD TEXT

Facilitation (ICF, SICF): Sensitivity analysis using meta::metacor()

```r
m.facilitation <- metacor(cor = cor,
              n = n,
              studlab = author,
              data = facilitation_r,
              cluster = cluster,
              common = FALSE,
              random = TRUE,
              method.tau = "REML",
              method.random.ci = "HK",
              title = "Sensitivity meta-analysis using metacor")

m.facilitation
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 8
## Number of estimates: k = 14
```

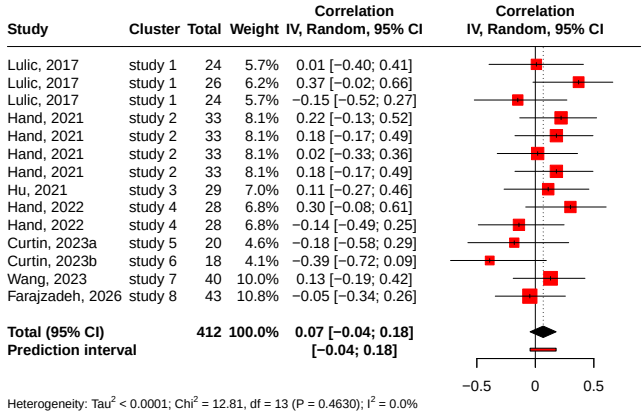
```
## Number of observations: o = 412
##
##                          COR            95%-CI    t p-value
## Random effects model 0.0673 [-0.0449; 0.1778] 1.30  0.2172
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0      [0.0000; 0.0458]; tau.1 = 0      [0.0000; 0.2141] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0463]; tau.2 < 0.0001 [0.0000; 0.2153] (within cluster)
##  I^2 = 0.0% [0.0%; 55.0%]; H = 1.00 [1.00; 1.49]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  12.81   13  0.4630
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 13)
## - Fisher's z transformation of correlations
```

#Cortical Facilitation

The multilevel random-effects meta-analysis indicated no statistically significant association between physical activity and cortical facilitation (r = 0.07, 95% CI [−0.04, 0.18], p = 0.22). The prediction interval ranged from −0.04 to 0.18, suggesting that the true effect in future studies is likely to be small and could range from slightly negative to moderately positive. No meaningful heterogeneity was observed across studies (I^2 = 0%), indicating that the variability in observed correlations was entirely attributable to sampling error. Variance decomposition confirmed that neither between-study nor within-study variance components contributed meaningfully to heterogeneity. Sensitivity analyses using a conventional random-effects model implemented with the metacor function yielded virtually identical results (r = 0.07, 95% CI [−0.04, 0.18], p = 0.22), supporting the robustness of the findings. To evaluate whether the association could be considered practically negligible, a two one-sided tests (TOST) equivalence test was conducted using equivalence bounds corresponding to a trivial effect size (−0.10 to 0.10). The equivalence test was not significant (Z = −0.63, p = 0.26), and the 90% confidence interval (−0.018 to 0.153) extended beyond the predefined equivalence bounds. Thus, equivalence could not be concluded. The conventional null hypothesis test was also non-significant (Z = 1.30, p = 0.20), indicating that the pooled association did not differ significantly from zero. Taken together, these results suggest that the available evidence is inconclusive regarding whether the true association is negligible or small. Subgroup analyses comparing intracortical facilitation measures (ICF vs. SICF) showed no significant difference between modulation types (Q = 0.00, df = 1, p = 0.97). Exploratory subgroup estimates were similar for ICF (r 0.07) and SICF (r 0.06).

Facilitation (ICF, SICF): Forest plot

```r
forest(m.facilitation,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Facilitation (ICF, SICF): Save plot

```
library(svglite)

svglite("forest_plot_facilitation.svg", height = 7, width = 7)

forest(m.facilitation,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Meta-analysis: Subgroup analyses

Inhibition

```
inhibition_r$inhibitory.measure <- as.factor(inhibition_r$inhibitory.measure)

inhibition_tms_subg <- rma.mv(yi = z,
                      V = var.z,
                      slab = author,
                      data = inhibition_r,
```

```r
                      random = ~ 1 | author/cor_id,
                      test = "t",
                      method = "REML",
                      mods = ~ inhibitory.measure)

summary(inhibition_tms_subg)
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   4.6056   -9.2113    2.7887    4.6042   30.7887
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000     11     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 10) = 5.7480, p-val = 0.8360
##
## Test of Moderators (coefficients 2:4):
## F(df1 = 3, df2 = 10) = 2.9454, p-val = 0.0851
##
## Model Results:
##
##                         estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt                   0.1438  0.1443   0.9961  10  0.3427  -0.1778  0.4654
## inhibitory.measureiSP     0.3602  0.1967   1.8311  10  0.0970  -0.0781  0.7984
## inhibitory.measureLICI   -0.1928  0.2141  -0.9007  10  0.3889  -0.6698  0.2842
## inhibitory.measureSICI   -0.0251  0.1569  -0.1600  10  0.8760  -0.3746  0.3244
##
## intrcpt
## inhibitory.measureiSP   .
## inhibitory.measureLICI
## inhibitory.measureSICI
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ADD TEXT

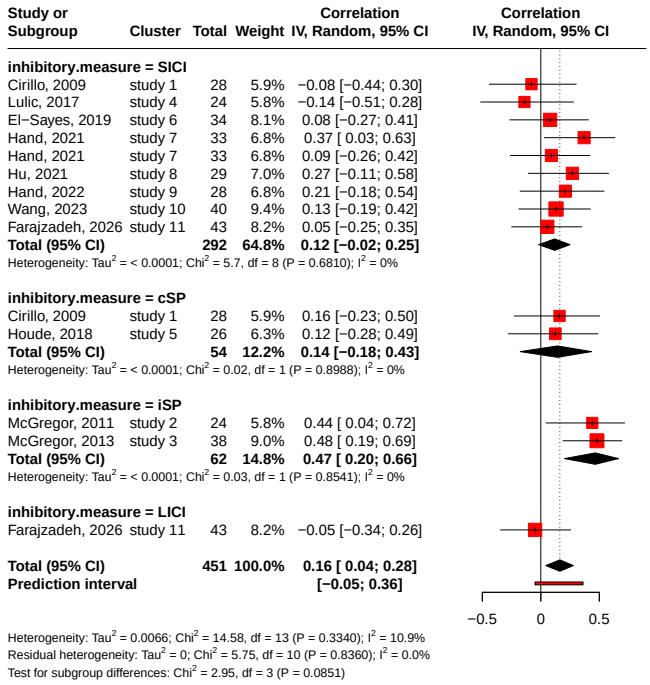
```r
inhibition_measure_type <- update(m.inhibition, subgroup = inhibitory.measure, tau.common = TRUE)
inhibition_measure_type
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 11
## Number of estimates: k = 14
## Number of observations: o = 451
##
```

```
##                          COR           95%-CI    t p-value
## Random effects model 0.1628 [0.0429; 0.2780] 2.92  0.0118
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0.0066 [0.0000; 0.0694]; tau.1 = 0.0814 [0.0000; 0.2634] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0572]; tau.2 < 0.0001 [0.0000; 0.2392] (within cluster)
##  I^2 = 10.9% [0.0%; 49.2%]; H = 1.06 [1.00; 1.40]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 = 0; tau = 0; I^2 = 0.0%; H = 1.00
##
## Test of heterogeneity:
##      Q d.f. p-value
##  14.58   13  0.3340
##
## Results for subgroups (random effects model):
##                              k     COR            95%-CI   tau^2     tau    Q
## inhibitory.measure = SICI   9  0.1181 [-0.0182; 0.2501] <0.0001 <0.0001 5.70
## inhibitory.measure = cSP    2  0.1428 [-0.1760; 0.4345] <0.0001 <0.0001 0.02
## inhibitory.measure = iSP    2  0.4652 [ 0.2033; 0.6650] <0.0001 <0.0001 0.03
## inhibitory.measure = LICI   1 -0.0490 [-0.3811; 0.2943] <0.0001 <0.0001 0.00
##                             I^2
## inhibitory.measure = SICI 0.0%
## inhibitory.measure = cSP  0.0%
## inhibitory.measure = iSP  0.0%
## inhibitory.measure = LICI   --
##
## Test for subgroup differences (random effects model):
##                   Q d.f. p-value
## Between groups 2.95    3  0.0851
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 13)
## - Fisher's z transformation of correlations
```

ADD TEXT

```r
forest(inhibition_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Inhibition measure type: Save plot

```
library(svglite)

svglite("forest_plot_inhibition-measure.svg", height = 10, width = 14)

forest(inhibition_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

Facilitation

```r
facilitation_r$facilitatory.measure <- as.factor(facilitation_r$facilitatory.measure)

facilitation_type <- rma.mv(yi = z,
                      V = var.z,
                      slab = author,
                      data = facilitation_r,
                      random = ~ 1 | author/cor_id,
                      test = "t",
                      method = "REML",
                      mods = ~ facilitatory.measure)

summary(facilitation_type)
```

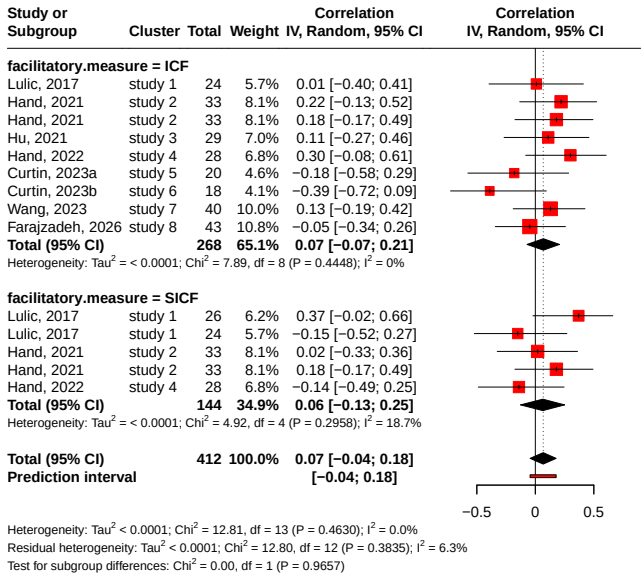
```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   1.9852   -3.9705    4.0295    5.9691    9.7438
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000      8     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 12.8033, p-val = 0.3835
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 0.0019, p-val = 0.9657
##
## Model Results:
##
##                            estimate      se     tval  df    pval    ci.lb
## intrcpt                      0.0691  0.0644   1.0727  12  0.3045  -0.0712
## facilitatory.measureSICF    -0.0048  0.1091  -0.0439  12  0.9657  -0.2425
##                             ci.ub
## intrcpt                    0.2094
## facilitatory.measureSICF   0.2329
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
facilitation_measure_type <- update(m.facilitation, subgroup = facilitatory.measure, tau.common = TRUE)
facilitation_measure_type
```

```
## Review:     Sensitivity meta-analysis using metacor
##
## Number of studies: n = 8
## Number of estimates: k = 14
## Number of observations: o = 412
##
```

```
##                          COR            95%-CI    t p-value
## Random effects model 0.0673 [-0.0449; 0.1778] 1.30  0.2172
##
## Quantifying heterogeneity (with 95%-CIs):
##  tau^2.1 = 0      [0.0000; 0.0458]; tau.1 = 0      [0.0000; 0.2141] (between cluster)
##  tau^2.2 < 0.0001 [0.0000; 0.0463]; tau.2 < 0.0001 [0.0000; 0.2153] (within cluster)
##  I^2 = 0.0% [0.0%; 55.0%]; H = 1.00 [1.00; 1.49]
##
## Quantifying residual heterogeneity (with 95%-CIs):
##  tau^2 < 0.0001; tau = 0.0017; I^2 = 6.3%; H = 1.03
##
## Test of heterogeneity:
##      Q d.f. p-value
##  12.81   13  0.4630
##
## Results for subgroups (random effects model):
##                               k    COR            95%-CI   tau^2     tau    Q
## facilitatory.measure = ICF    9 0.0690 [-0.0711; 0.2064] <0.0001 <0.0001 7.89
## facilitatory.measure = SICF   5 0.0642 [-0.1268; 0.2507] <0.0001 <0.0001 4.92
##                               I^2
## facilitatory.measure = ICF   0.0%
## facilitatory.measure = SICF 18.7%
##
## Test for subgroup differences (random effects model):
##                   Q d.f. p-value
## Between groups 0.00    1  0.9657
##
## Details of meta-analysis methods:
## - Inverse variance method (three-level model)
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Profile-Likelihood method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Random effects confidence interval based on t-distribution (df = 13)
## - Fisher's z transformation of correlations
```

```r
forest(facilitation_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)
```



Facilitation by tms measure type: Save plot

```
library(svglite)

svglite("forest_plot_facilitation-type.svg", height = 7, width = 14)

forest(facilitation_measure_type,
       layout = "Revman5",
       common = FALSE,
       xlim = c(-0.9,0.9),
       prediction = TRUE,
       fs.hetstat = 10,
       col.subgroup = "black",
       addrows.below.overall = 2)

dev.off()
```

```
## pdf
##   2
```

#Meta Regressions ## Excitability ### Meta-regressions by proportion of women

```
m.cor.reg.excitability.women <- metareg(m.excitability, ~Prop_women)
m.cor.reg.excitability.women
```

```
##
## Multivariate Meta-Analysis Model (k = 15; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000     14     no        .id
```

```
## sigma^2.2  0.0054  0.0733     15     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 13) = 17.5691, p-val = 0.1746
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 13) = 0.8025, p-val = 0.3866
##
## Model Results:
##
##             estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt       0.1420  0.1743   0.8148  13  0.4299  -0.2345  0.5185
## Prop_women   -0.2752  0.3072  -0.8958  13  0.3866  -0.9388  0.3884
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by age

```r
m.cor.reg.excitability.age <- metareg(m.excitability, ~Age)
m.cor.reg.excitability.age
```

```
##
## Multivariate Meta-Analysis Model (k = 15; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000     14     no       .id
## sigma^2.2  0.0063  0.0793     15     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 13) = 17.4034, p-val = 0.1815
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 13) = 0.9785, p-val = 0.3406
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt   -0.1060  0.1149  -0.9228  13  0.3729  -0.3541  0.1421
## Age        0.0026  0.0027   0.9892  13  0.3406  -0.0031  0.0084
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by study quality

```r
m.cor.reg.excitability.quality <- metareg(m.excitability, ~Q_score)
m.cor.reg.excitability.quality
```

```
##
## Multivariate Meta-Analysis Model (k = 15; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000     14     no       .id
## sigma^2.2  0.0097  0.0983     15     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 13) = 18.3808, p-val = 0.1436
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 13) = 0.1239, p-val = 0.7305
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt   -0.1691  0.4663  -0.3627  13  0.7227  -1.1764  0.8382
## Q_score    0.0121  0.0345   0.3520  13  0.7305  -0.0624  0.0867
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Inhibition

Meta-regressions by proportion of women

```r
m.cor.reg.inhibition.women <- metareg(m.inhibition, ~Prop_women)
m.cor.reg.inhibition.women
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0086  0.0925     11     no       .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 14.5714, p-val = 0.2657
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 0.0000, p-val = 0.9969
##
## Model Results:
##
##             estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt       0.1655  0.0621   2.6664  12  0.0205   0.0303  0.3007  *
## Prop_women   -0.0000  0.0035  -0.0040  12  0.9969  -0.0077  0.0076
##
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by age

```r
m.cor.reg.inhibition.age <- metareg(m.inhibition, ~Age)
m.cor.reg.inhibition.age
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0011  0.0339     11     no       .id
## sigma^2.2  0.0000  0.0001     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 11.1893, p-val = 0.5128
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 3.2645, p-val = 0.0959
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt   -0.0434  0.1232  -0.3526  12  0.7305  -0.3119  0.2250
## Age        0.0060  0.0033   1.8068  12  0.0959  -0.0012  0.0133  .
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by study quality

```r
m.cor.reg.inhibition.quality <- metareg(m.inhibition, ~Q_score)
m.cor.reg.inhibition.quality
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0050  0.0704     11     no       .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 11.4779, p-val = 0.4885
##
```

```
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 2.8514, p-val = 0.1171
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt   -0.6956  0.5114  -1.3601  12  0.1988  -1.8098  0.4187
## Q_score    0.0659  0.0390   1.6886  12  0.1171  -0.0191  0.1508
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Facilitation

Meta-regressions by proportion of women

```r
m.cor.reg.facilitation.women <- metareg(m.facilitation, ~Prop_women)
m.cor.reg.facilitation.women
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000      8     no        .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 11.8632, p-val = 0.4567
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 0.9420, p-val = 0.3509
##
## Model Results:
##
##             estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt       0.1652  0.1134   1.4572  12  0.1707  -0.0818  0.4123
## Prop_women   -0.2199  0.2265  -0.9706  12  0.3509  -0.7134  0.2737
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by age

```r
m.cor.reg.facilitation.age <- metareg(m.facilitation, ~Age)
m.cor.reg.facilitation.age
```

```
##
```

```
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000      8     no        .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 10.4762, p-val = 0.5743
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 2.3291, p-val = 0.1529
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    0.3206  0.1738   1.8442  12  0.0900  -0.0582  0.6994  .
## Age       -0.0089  0.0059  -1.5261  12  0.1529  -0.0217  0.0038
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Meta-regressions by study quality

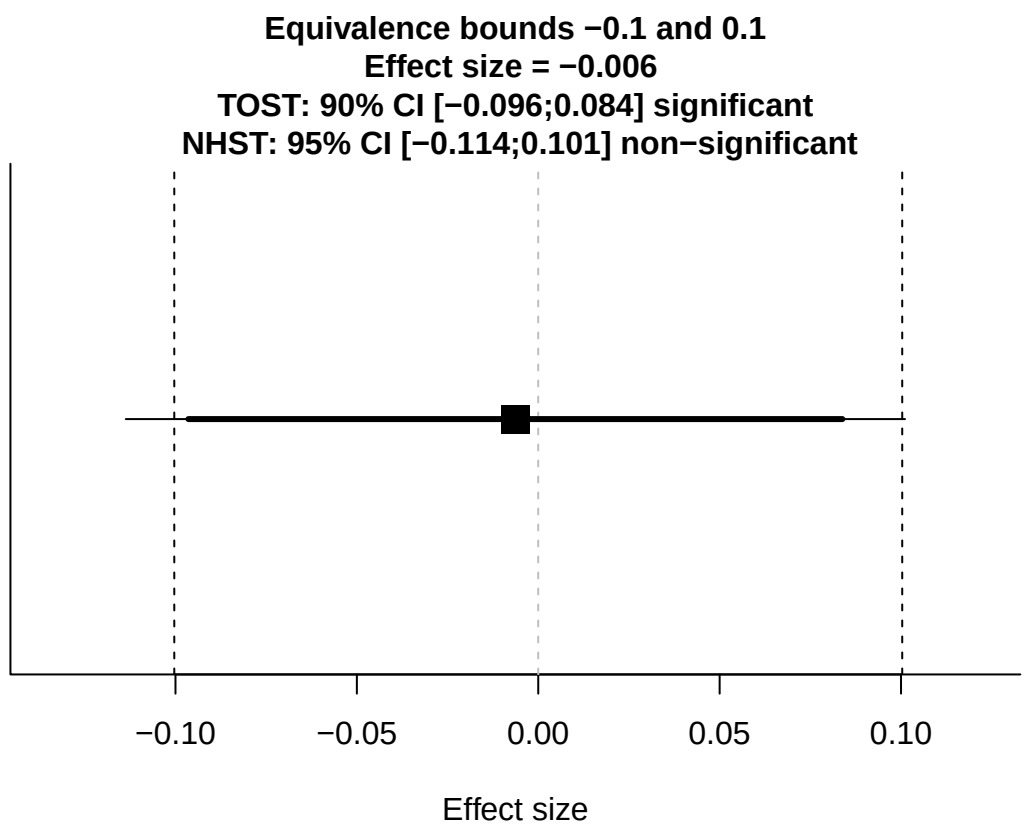
```r
m.cor.reg.facilitation.quality <- metareg(m.facilitation, ~Q_score)
m.cor.reg.facilitation.quality
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed    factor
## sigma^2.1  0.0000  0.0000      8     no        .id
## sigma^2.2  0.0000  0.0000     14     no  .id/.idx
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 9.7305, p-val = 0.6396
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 3.0747, p-val = 0.1050
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    0.7979  0.4198   1.9006  12  0.0816  -0.1168  1.7126  .
## Q_score   -0.0571  0.0326  -1.7535  12  0.1050  -0.1281  0.0139
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Equivalence Testing

Excitability

```r
# Define SESOI in r
sesoi_r <- 0.1

# Convert r to Fisher's z
low_eq_z  <- atanh(-sesoi_r)
high_eq_z <- atanh( sesoi_r)

TOSTmeta(
  ES = -0.0063,
  se = 0.0548,
  low_eqbound  = low_eq_z,
  high_eqbound =  high_eq_z,
  alpha = 0.05
)
```

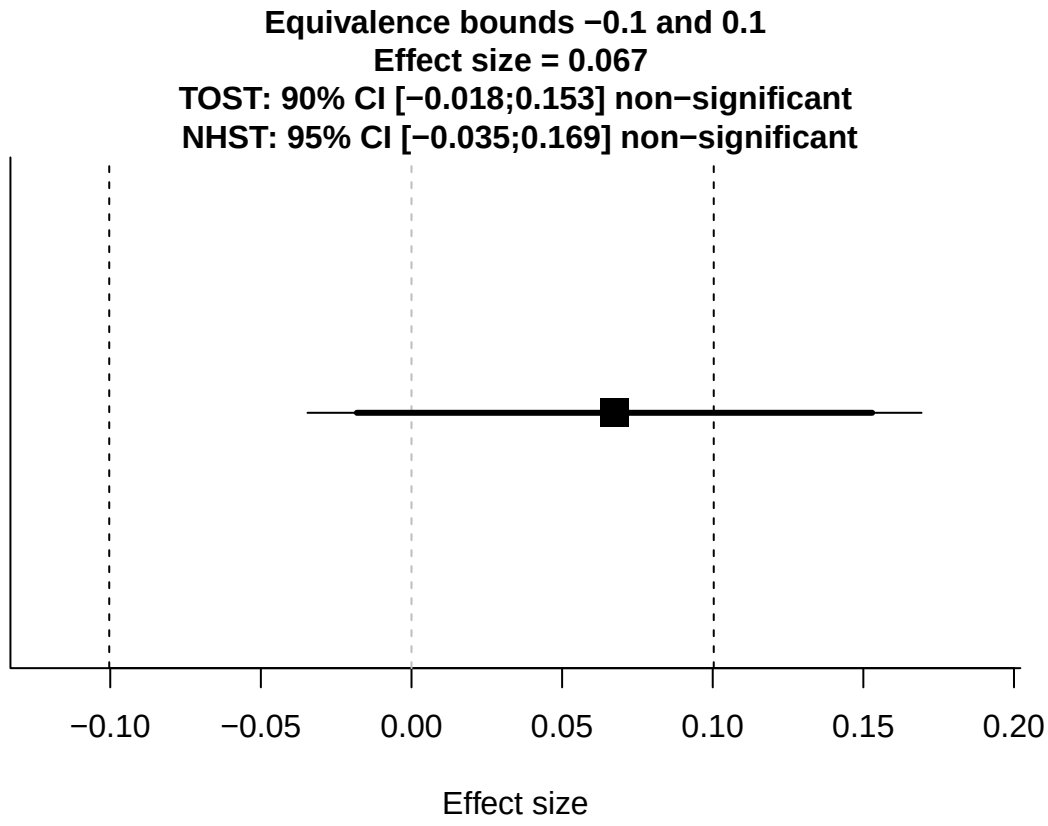



```
## TOST results:
## Z-value lower bound: 1.72    p-value lower bound: 0.043
## Z-value upper bound: -1.95   p-value upper bound: 0.026
##
## Equivalence bounds (Cohen's d):
```

```
## low eqbound: -0.1003
## high eqbound: 0.1003
##
## TOST confidence interval:
## lower bound 90% CI: -0.096
## upper bound 90% CI:  0.084
##
## NHST confidence interval:
## lower bound 95% CI: -0.114
## upper bound 95% CI:  0.101
##
## Equivalence Test Result:
## The equivalence test was significant, Z = 1.716, p = 0.0431, given equivalence bounds of -0.100 and 0

##

##
## Null Hypothesis Test Result:
## The null hypothesis test was non-significant, Z = -0.115, p = 0.908, given an alpha of 0.05.

##

## NHST: don't reject null significance hypothesis that the effect is equal to 0
## TOST: reject null equivalence hypothesis
```

Facilitation

```r
# Define SESOI in r
sesoi_r <- 0.1

# Convert r to Fisher's z
low_eq_z  <- atanh(-sesoi_r)
high_eq_z <- atanh( sesoi_r)

TOSTmeta(
  ES = 0.0674,
  se = 0.0520,
  low_eqbound  = low_eq_z,
  high_eqbound =  high_eq_z,
  alpha = 0.05
)
```




```
## TOST results:
## Z-value lower bound: 3.23    p-value lower bound: 0.0006
## Z-value upper bound: -0.633  p-value upper bound: 0.263
##
## Equivalence bounds (Cohen's d):
## low eqbound: -0.1003
## high eqbound: 0.1003
##
## TOST confidence interval:
## lower bound 90% CI: -0.018
## upper bound 90% CI:  0.153
##
## NHST confidence interval:
## lower bound 95% CI: -0.035
## upper bound 95% CI:  0.169
##
## Equivalence Test Result:
## The equivalence test was non-significant, Z = -0.633, p = 0.263, given equivalence bounds of -0.100 a

##

##
## Null Hypothesis Test Result:
## The null hypothesis test was non-significant, Z = 1.296, p = 0.195, given an alpha of 0.05.

##
```

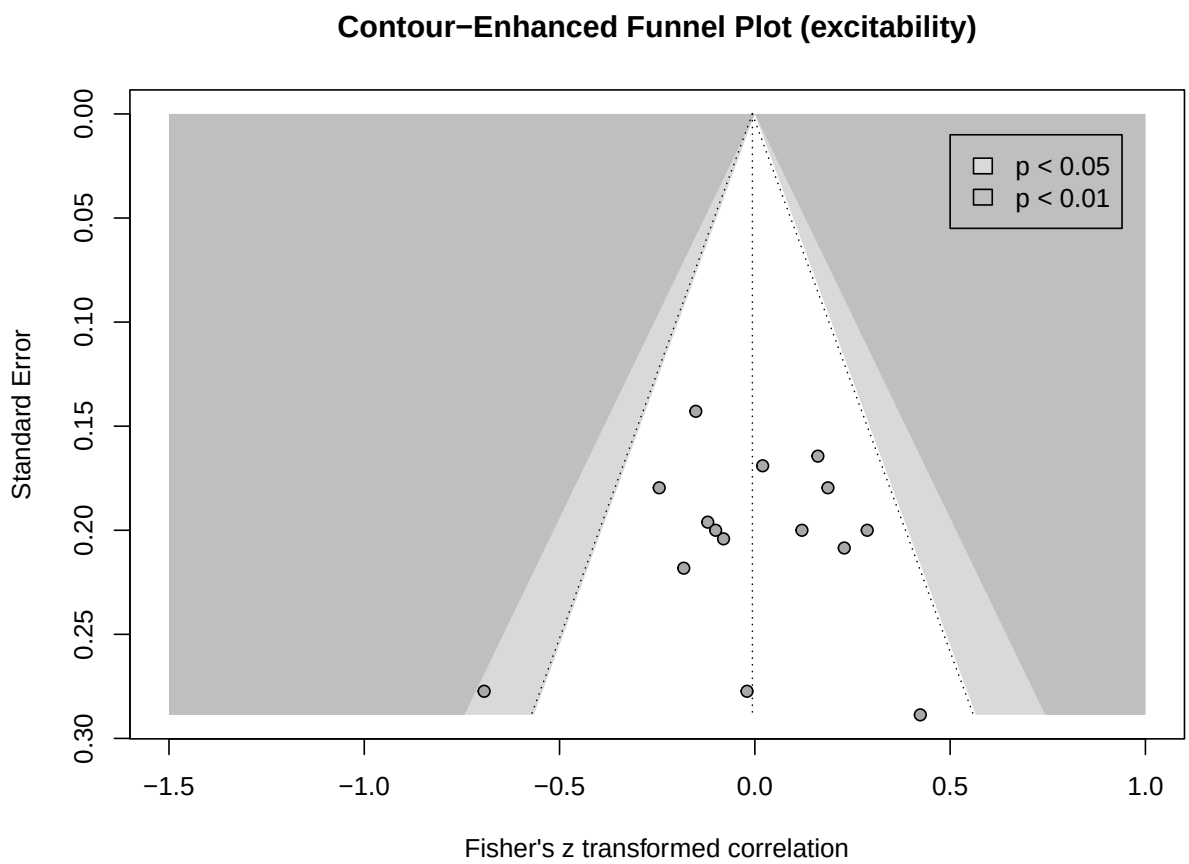
```
## NHST: don't reject null significance hypothesis that the effect is equal to 0
## TOST: don't reject null equivalence hypothesis
```

Publication bias analysis

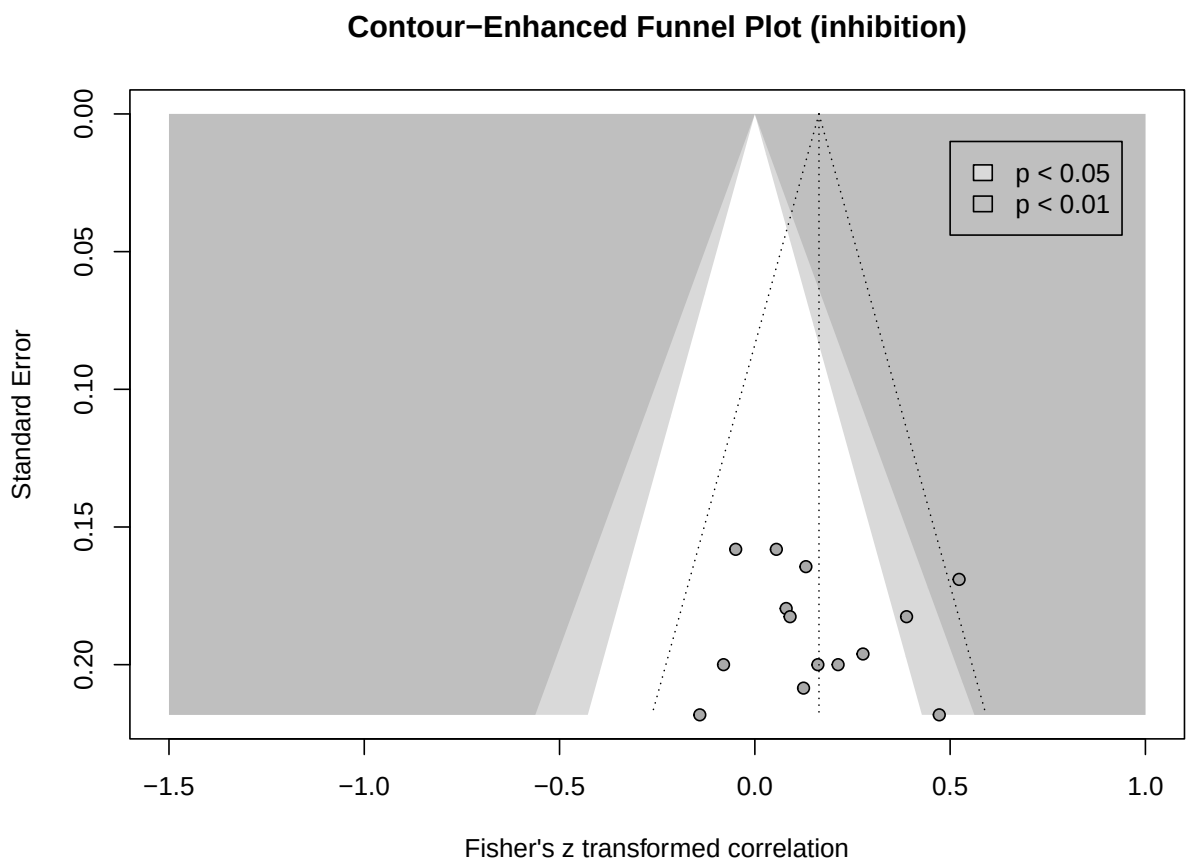
Small-study effects

Funnel plot

```r
# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.excitability,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p < 0.05", "p < 0.01"),
       fill = col.contour)

# title
title("Contour-Enhanced Funnel Plot (excitability)")
```



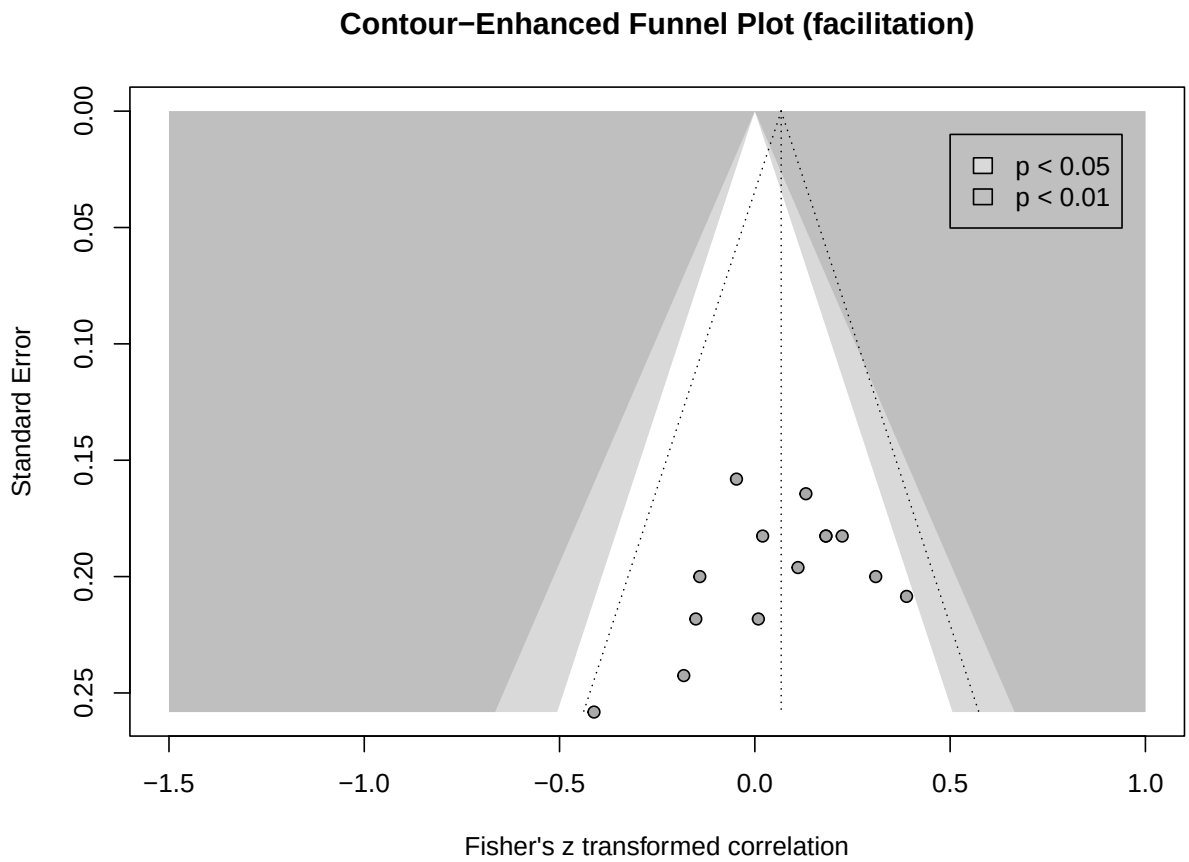

```r
# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.inhibition,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p < 0.05", "p < 0.01"),
       fill = col.contour)

# title
title("Contour-Enhanced Funnel Plot (inhibition)")
```




```r
png(file = "Funnel Plot.png", width = 2500, height = 2000, res = 300)

# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.inhibition,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p < 0.05", "p < 0.01"),
       fill = col.contour)

# title
title("Contour-Enhanced Funnel Plot (tms and Physical Activity)")

dev.off()
```

```
## pdf
##   2
```

```r
# Define fill colors for contour
col.contour = c("gray85", "gray75")

# Funnel plot
meta::funnel(m.facilitation,
             xlim = c(-1.5, 1),
             contour = c(0.95, 0.99),
             col.contour = col.contour)

# legend
legend(x =0.5, y = 0.01,
       legend = c("p < 0.05", "p < 0.01"),
       fill = col.contour)

# title
title("Contour-Enhanced Funnel Plot (facilitation)")
```



Egger's test

There is no direct function to conduct Egger's test for multi-level model. Alternatively, we calculate it by using the standard errors of the effect size estimates as a predictor in the meta-regression. (see: https://rstudio-pubs-static.s3.amazonaws.com/814435_6401e0cceb0b410c8208642dc5ee07f0.html)

```r
test.egger = rma.mv(z,var.z, mod = ~ sqrt(var.z), random =  ~ 1 | author/cor_id, data = excitability_r,
test.egger
```

```
##
## Multivariate Meta-Analysis Model (k = 15; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000     14     no         author
## sigma^2.2  0.0099  0.0997     15     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 13) = 18.5318, p-val = 0.1384
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 13) = 0.0340, p-val = 0.8566
##
## Model Results:
##
##              estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt        0.0500  0.3105   0.1611  13  0.8745  -0.6208  0.7209
## sqrt(var.z)   -0.2879  1.5622  -0.1843  13  0.8566  -3.6628  3.0870
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
test.egger = rma.mv(z,var.z, mod = ~ sqrt(var.z), random =  ~ 1 | author/cor_id, data = inhibition_r, t
test.egger
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0102  0.1008     11     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 14.5085, p-val = 0.2694
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 0.0072, p-val = 0.9340
##
## Model Results:
##
##              estimate      se    tval  df    pval    ci.lb   ci.ub
## intrcpt        0.1196  0.5547  0.2156  12  0.8329  -1.0890  1.3282
## sqrt(var.z)    0.2509  2.9674  0.0846  12  0.9340  -6.2146  6.7164
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
test.egger = rma.mv(z,var.z, mod = ~ sqrt(var.z), random =  ~ 1 | author/cor_id, data = facilitation_r,
test.egger
```

```
##
## Multivariate Meta-Analysis Model (k = 14; method: REML)
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed         factor
## sigma^2.1  0.0000  0.0000      8     no         author
## sigma^2.2  0.0000  0.0000     14     no  author/cor_id
##
## Test for Residual Heterogeneity:
## QE(df = 12) = 10.6091, p-val = 0.5627
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 12) = 2.1961, p-val = 0.1641
##
## Model Results:
##
##              estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt        0.6604  0.4035   1.6367  12  0.1276  -0.2187  1.5395
## sqrt(var.z)   -3.0740  2.0743  -1.4819  12  0.1641  -7.5934  1.4455
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Pulication Bias Egger's regression tests using the standard errors of the effect size estimates as a predictor in the meta-regression showed that the coefficient of the standard error was not significant for any of the metaanalyses ($p > 0.164$), confirming that the data in the funnel plot was symmetrical, suggesting the absence of publication bias.

References

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